

UNIVERSITY OF THE
PACIFIC
Thomas J. Long School of
Pharmacy & Health Sciences

Re: Zak Raymond
01/02/2025

Atefeh Rabiee, Ph.D.
Assistant Professor
Dept. of Pharmaceutical Sciences
Thomas J. Long School of Pharmacy,
University of the Pacific
751 Brookside Rd, Stockton CA 95211
Email: arabiee@pacific.e
Phone: +1 209-946-7325

Dear REU Selection Committee,

I am writing with great enthusiasm to recommend Zakary Raymond for your Research Experiences for Undergraduates (REU) program. As Zakary's research mentor at the University of the Pacific, I have directly supervised his work on adipose stem cell research and lipedema studies since February 2024, and I can confidently say that he is one of the most promising undergraduate researchers I have worked with.

Zakary has demonstrated exceptional aptitude for laboratory research through his involvement in multiple projects in my lab. His most significant contribution has been his work on optimizing adipogenic differentiation protocols for lipedema-derived adipose stem cells. This complex project required mastery of various specialized techniques, including cell culture, western blotting, qPCR, and Oil Red O staining. What particularly impressed me was Zakary's ability to independently troubleshoot technical challenges and his methodical approach to protocol optimization. The quality of his research was recently recognized when he was selected to present at the 2025 National Collegiate Research Conference at Harvard University.

Beyond his technical skills, Zakary possesses the intellectual curiosity and analytical mindset essential for successful research. He consistently demonstrates initiative in exploring scientific literature, proposing thoughtful experimental modifications, and connecting his findings to broader implications for lipedema treatment. His work has contributed valuable insights into the metabolic characteristics of lipedema-derived stem cells, which may help advance our understanding of this understudied condition affecting millions of women worldwide.

Zakary's strong academic performance (3.87 GPA) in a challenging biomedical engineering curriculum, combined with his research experience, makes him exceptionally well-prepared for an intensive research program. He has already proven his ability to balance multiple responsibilities, as evidenced by his concurrent roles as a Residential Assistant and involvement in various leadership positions while maintaining his research commitments.

What sets Zakary apart is his genuine passion for biomedical research and his ability to work both independently and collaboratively. He has shown remarkable growth in his scientific thinking and technical capabilities during his time in my lab. His recent work on developing western blotting protocols for quantifying α -SMA in corneal cells demonstrates his versatility and ability to quickly adapt to new research areas.

I am confident that Zakary would make exceptional contributions to your REU program. He has the intellectual capacity, laboratory skills, and professional maturity to excel in an intensive

research environment. His demonstrated ability to conduct independent research while collaborating effectively with others makes him an ideal candidate for this opportunity.

Please feel free to contact me if you require any additional information about Zakary's qualifications or his research experience in my laboratory.

Cordially,

A handwritten signature in cursive script that reads "Atefeh Rabiee".

Dr. Atefeh Rabiee
Thomas J. Long School of Pharmacy
University of the Pacific